

Problem 1 – Question 3

Implementing $R_{\mathbf{n}}(\theta)$ with R_Y and R_Z

Parameterize the unit vector $\mathbf{n}$ by polar and azimuthal angles:

$$\mathbf{n} = (\sin \lambda \cos \phi, \sin \lambda \sin \phi, \cos \lambda).$$

Let

$$W = R_Z(\phi)R_Y(\lambda).$$

Rotations conjugate Pauli vectors in the same way that their corresponding three-dimensional rotations act on ordinary vectors. In particular,

$$\begin{aligned} WZW^\dagger &= R_Z(\phi)(R_Y(\lambda)ZR_Y(-\lambda))R_Z(-\phi) \\ &= \sin \lambda \cos \phi X + \sin \lambda \sin \phi Y + \cos \lambda Z \\ &= \mathbf{n} \cdot \boldsymbol{\sigma}. \end{aligned}$$

Therefore

$$\begin{aligned} R_{\mathbf{n}}(\theta) &= \exp\left(-i\frac{\theta}{2}WZW^\dagger\right) \\ &= W \exp\left(-i\frac{\theta}{2}Z\right)W^\dagger \\ &= R_Z(\phi)R_Y(\lambda)R_Z(\theta)R_Y(-\lambda)R_Z(-\phi). \end{aligned}$$

Thus a valid gate sequence, listed in chronological order, is

$$\boxed{R_Z(-\phi), \quad R_Y(-\lambda), \quad R_Z(\theta), \quad R_Y(\lambda), \quad R_Z(\phi)}.$$

The rightmost factor in the matrix product acts first, which explains the reversal between the product and the chronological list.

Special cases. If $\mathbf{n} = \hat{z}$, take $\lambda = 0$, and the construction reduces to $R_Z(\theta)$. If $\mathbf{n} = -\hat{z}$, take $\lambda = \pi$. At either pole, ϕ is arbitrary and may be set to zero.

Resource count. The direct construction uses five elementary rotations. Adjacent rotations about the same axis can be merged with neighboring circuit gates. In a larger circuit this often reduces the effective count; for example, $R_Z(a)R_Z(b) = R_Z(a + b)$.